

Assignment no.02

Bt-501

Described known chromosomal disorders and then indicate type of chromosomal abnormalities and syndromes that arise each type of chromosomal disorders.

Chromosomal disorders:



- **A chromosome disorder results from a change in the number or structure of chromosomes.**
- Each of our chromosomes[?] has a characteristic structure.
- Historically, scientists have used a staining technique that colours the chromosomes into a banding pattern.
- These banding patterns make each of our individual chromosomes easier to identify, like a map.
- A set of chromosomes, as seen under a microscope, is known as a karyotype[?].
- Any deviation from the normal karyotype is known as a chromosome abnormality.
- While some chromosome abnormalities are harmless, some are associated with clinical disorders.
- Half of all spontaneous abortions are due to chromosome abnormalities.

Numerical abnormalities

- The most severe chromosome disorders are caused by the loss or gain of whole chromosomes, which can affect hundreds, or even thousands, of genes[?] and are usually fatal.
- A few numerical abnormalities support development to term, either because the chromosome is small and/or contains relatively few genes or because there is a natural mechanism present to help adjust gene dosage.
- The major numerical abnormalities that survive to term are presented in Table 1, with Down's syndrome[?] being the most common.

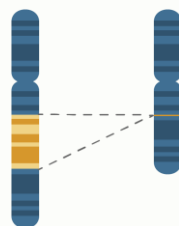
The major numerical abnormalities that survive to term

Syndrome	Abnormality	Incidence
Down's	Trisomy 21	15 in 10,000
Edwards'	Trisomy 18	3 in 10,000
Patau's	Trisomy 13	2 in 10,000
Turner	Monosomy X	2 in 10,000 (female births)
Klinefelter's	XXY	10 in 10,000 (male births)
XXX	XXX	10 in 10,000 (female births)
XYY	XYY	10 in 10,000 (male births)

Structural abnormalities

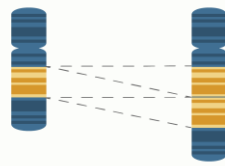
- This is when large sections of DNA[?] are missing from or are added to a chromosome.
- Structural abnormalities can take several forms.
- **Deletion:** a mutation[?] causing part of the chromosome to be missing.

Deletion



- **Duplication:** a mutation causing part of the chromosome to be repeated, resulting in extra genetic material.

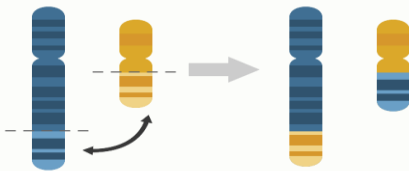
Duplication



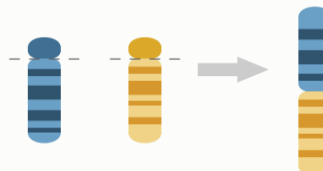
- **Translocation:** a mutation causing one portion of a chromosome to be moved to a different part of the chromosome (intrachromosomal) or to a different chromosome altogether (interchromosomal). There are two key types:

- reciprocal: segments from two different chromosomes are exchanged
- Robertsonian: an entire chromosome attaches to another.

Reciprocal translocation

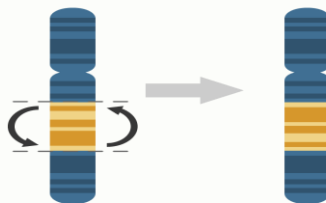


Robertsonian translocation



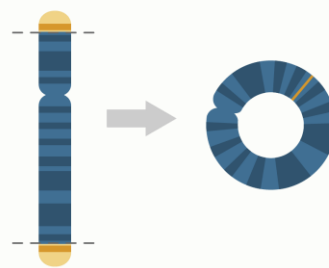
- **Inversion:** a mutation resulting in a portion of a chromosome being in the opposite orientation (inverted).

Inversion



- **Ring:** when a portion of a chromosome has broken off and formed a circle or ring.

Ring chromosome



Balanced vs unbalanced structural abnormalities

- Balanced structural abnormalities involve the rearrangement of genetic material but with no overall gain or loss. For example, inversions and translocations.
- Balanced structural abnormalities can have an immediate effect, but the major consequence is the production of eggs or sperm with incomplete or partially duplicated sets of chromosomes.
- Unbalanced structural abnormalities involve genetic material being gained or lost.
- Even tiny unbalanced structural abnormalities can affect many genes and, consequently, have severe effects on the individual – as listed in the table below.

Table 2: Unbalanced structural abnormalities

Syndrome	Abnormality	Incidence	
Wolf-Hirschhorn	Deletion from tip of short arm of chromosome 4	1 in 50,000	
Cri-du-chat	Deletion from tip of short arm of chromosome 5	1 in 50,000	
WAGR syndrome	Microdeletion from short arm of chromosome 11	1 in 500,000 to 1 million	
Prader-Willi/Angelman	Microdeletion from short arm of chromosome 15	1 in 15,000	
DiGeorge	Microdeletion from long arm of chromosome 22	1 in 4,000	

How do chromosome abnormalities occur?

- Chromosome abnormalities usually occur when there is an error in cell division resulting in cells with too few or too many copies of a chromosome.
- Most chromosome abnormalities originate in the egg or sperm (gametes) but some happen during embryo development or are inherited from a parent.
- Chromosome abnormalities that originate in the gametes are present in every cell of the body.
- Normally during the formation of gametes the two pairs of chromosomes (one from the mother and one from the father), separate in a process called meiosis. This results in just one copy of each chromosome in the gametes (as opposed to the two copies found in the other cells of the body).
- Errors in this separation process lead to the formation of gametes with incomplete sets of chromosomes or additional whole or parts of chromosomes. This is known as meiotic nondisjunction.
- Other factors that can increase the risk of chromosome abnormalities include maternal age (the frequency of meiotic nondisjunction increases with maternal age) and environmental factors such as exposure to certain drugs.

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